PROJECT PROPOSAL

Date of proposal:
15-Sep 2026

Project Title:
MealSenseAI

Group ID (As Enrolled in Canvas Class Groups):
Practice Project Group (PT& Stackable) 31

Group Members (name , Student ID):
Dizon Joselito Villanueva, A0362372E
Individual Project Submission

Sponsor/Client: *(Company Name, Address and Contact Name, Email, if any)*
This project will be delivered without any sponsorship.
The use cases are based on personal needs and perhaps useful for public as well.

Background/Aims/Objectives:

People often struggle to identify what is on their plate, estimate its nutritional content accurately, and understand its short- and long-term health effects.

The proposed solution is an intelligent meal analysis system that helps:

1. *Identifies food and beverages from a camera view.*
2. *Estimates the volume and portion size of each item.*
3. *Determines the meal's nutritional content.*
4. *Analyses potential health effects against dynamic personal baselines.*
5. *Provides insights aligned with user health goals, such as general wellness, diabetes management, weight loss, weight gain, or endurance training.*

The aim is to provide an intuitive solution that delivers actionable information to help people make informed decisions.

Project Description:

MealSenseAI is an intelligent meal recording system that uses computer vision and pattern recognition to identify food items from images, estimate portions, and calculate nutritional content.

The application analyses meals using computer vision and neural network, then combines the results with the user's health profile, dietary goals, and eating patterns to provide personalised insights and recommendations.

The initial focus for this MVP is supporting people with Type 1 and Type 2 diabetes, with future applications for endurance, performance, and lifestyle nutrition management.

Project Background / Market Context:

After Singapore's COVID-19 Circuit Breaker, I was diagnosed with Type 2 diabetes, the first in my family. It made me realize the difficulty to understand what I eat, its nutritional content, and how everyday food choices affect my health.

My experience reflects a much larger and unequal problem. WHO estimates that 830 million people were living with diabetes in 2022 (WHO, 2024)*, with prevalence rising faster in low- and middle-income countries* (WHO, 2024)*. Access to treatment is also unequal, more than half of adults aged 30 and above with diabetes were not taking medication in 2022, with the lowest treatment coverage in low- and middle-income countries* (WHO, 2024)*.*

In Singapore, Diabetes prevalence remains around 8.5% (MOH, 2023)*, while obesity increased from 8.6% in 2013 to 11.6% in 2021–2022* (MOH, 2023)*. Without effective prevention, Singapore has previously projected that as many as one million residents could be living with diabetes by 2050* (MOH, 2017)*.*

Managing diabetes and obesity requires people to understand what they eat, particularly carbohydrates, sugar, calories and portion size. Yet nutritional information is often difficult to interpret or unavailable, especially for unpackaged, home-cooked and street food. This creates a potential food information gap or social divide. People with greater access to dietitians, health services, education and digital tools are better positioned to make informed dietary decisions (MOH, 2026)*.*

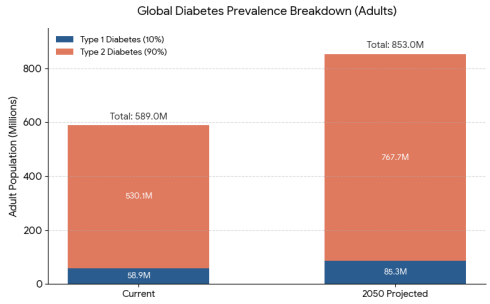
Evidence shows that simpler information can change behaviour. Singapore's Nutri-Grade initiative provides a useful example. After implementation, household sugar purchases from pre-packaged beverages fell by approximately 18%, while purchases of less-healthy beverages declined by 44% (Shin & Tan, 2026)*. Singapore's Nutri-Grade initiative demonstrates that simpler and more visible nutritional information can influence healthier choices. However, this information is largely available for labelled products and beverages, leaving a gap when people consume mixed, home-cooked, restaurant, or unlabelled meals.*

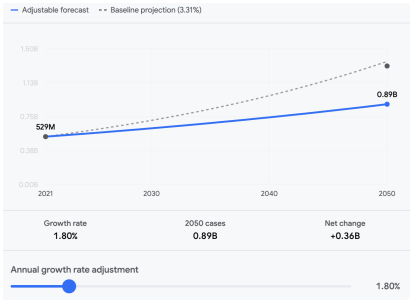
To address the gap, MealSenseAI is an AI-enabled meal analysis system that brings nutritional intelligence directly to the plate. Using an ordinary smartphone camera, it could identify food, estimate portions and nutritional content, and translate them into simple, personalised guidance based on the user's health goals.

Most importantly, a scalable, low-cost solution could extend basic nutritional intelligence to communities with fewer healthcare resources. This supports the UN Sustainable Development Goal 3, particularly improving equitable access to healthcare turning AI-assisted nutrition from a convenience into a potential tool for reducing health inequality (UN, 2026)*.*

Market Research - Opportunity

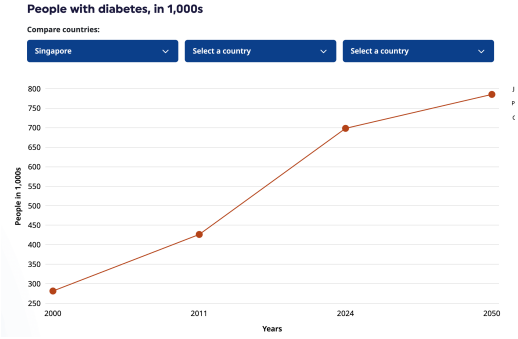
Diabetes affects approximately 589 million adults globally, with more than 90% having Type 2 diabetes; this is projected to reach 853 million by 2050, at a growth of 1.8% year on year. Type 1 diabetes represents a smaller but highly relevant segment because daily management depends heavily on glucose monitoring, insulin dosing and carbohydrate estimation (IDF, 2025).

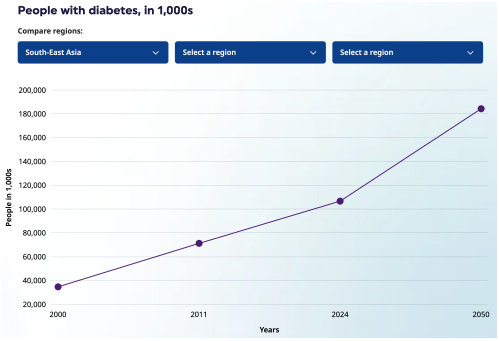




Southeast Asian Context

In Singapore latest national survey, reports diabetes prevalence at 9.1%, while MOH states that 1 in 3 Singaporeans is at risk of developing diabetes during their lifetime. IDF separately estimates approximately 699,100 adults with diabetes in Singapore in 2024 (MOH, 2025).





Diabetes represents a large and growing opportunity across Southeast Asia. The broader South-East Asia Region recorded an increase from 34.9 million adults with diabetes in 2000 to 106.9 million in 2024, and is projected to reach 184.5 million by 2050 (IDF, 2025).

Techno-Economic Analysis (TEA)

(Using a phased approach Singapore -> Southeast Asia subscription model.)

Phase 1 – Singapore: *With approximately 699,100 adults living with diabetes, an initial 1% market penetration (~6,991 users) at SGD 3.99/month could generate approximately SGD 27,894 in initial monthly gross revenue. Assuming the customer base grows by 1% monthly, Year 1 gross revenue is estimated at SGD 353,767.*

Phase 2 – Southeast Asia: *Regional expansion significantly increases the addressable market. Using a lowered SGD 2.99/month subscription and the same 1% monthly customer growth assumption, the model can scale progressively from initial market entry toward millions of subscribers. At 3.9 million paying users, the scale scenario could generate approximately SGD 147.89 million in Year 1 gross revenue.*

For preliminary TEA, operating costs including cloud and AI infrastructure, engineering, platform operations, marketing and administration are assumed at 40–60% of revenue, with 50% used as the base-case planning assumption.

Scenario	*Paying Users*	*Price / Month*	*Year 1 Gross Revenue*	*Base OpEx (50%)*	*Indicative Operating Profit*
Singapore – 1% penetration	*6,991*	*SGD 3.99*	*~SGD 353.8K*	*~SGD 176.9K*	*~SGD 176.9K*
SEA – Scale Scenario	*3.9M*	*SGD 2.99*	*~SGD 147.9M*	*~SGD 73.9M*	*~SGD 73.9M*

These figures are illustrative market scenarios rather than financial forecasts. They demonstrate that MealSenseAI can begin with a focused Singapore market, validate product-market fit, and progressively scale across Southeast Asia while benefiting from the economics of a digital subscription platform.

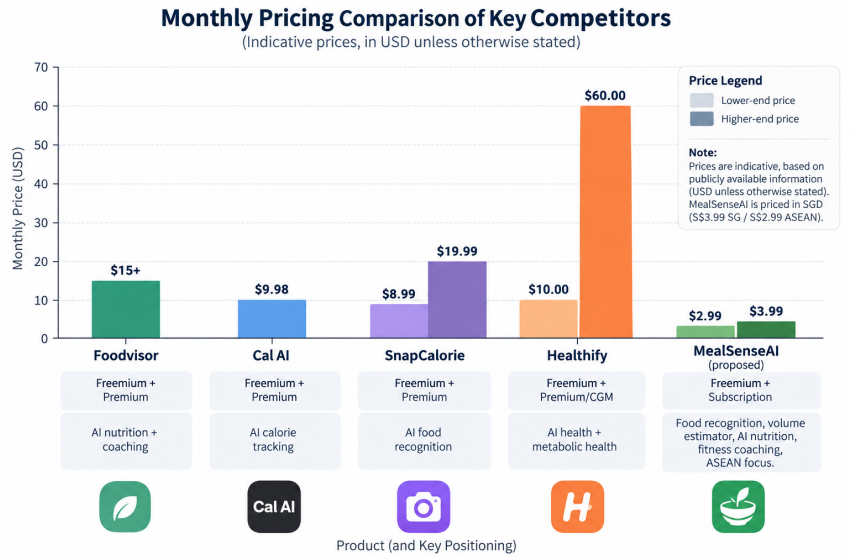
Competitive Landscape and Pricing Analysis

MealSenseAI is entering an established market and is not a novel concept. However, advances in AI are levelling the playing field (Kenny, Kowalkiewicz, & Oosthuizen, 2024). *Competitive advantage will depend on how quickly products adopt AI, apply AI to real business problems and build differentiated capabilities around it* (Diedrich, Williams, Catlin, & Fountaine, 2025).

The goal is not to be the cheapest, but to deliver the right features around real user needs at a competitive price encouraging users to make proactive and informed health choices.

Product	*Model*	*Indicative Price*	*Positioning*
Foodvisor	*Freemium + Premium*	*US$15+/month*	*AI nutrition + coaching*
Cal AI	*Freemium + Premium*	*US$9.98/month*	*AI calorie tracking*
SnapCalorie	*Freemium + Premium*	*US$8.99-$19.99/Month*	*AI food recognition*
Healthify	*Freemium + Premium/CGM*	*US$10.00-$60.00/Month*	*AI health + metabolic health*

MealSenseAI proposed	*Freemium + Subscription*	*S$3.99 SG / S$2.99 ASEAN*	*Food Recognition, Volume estimator, AI Nutrition, Fitness Coaching, ASEAN focus.*



Project Scope

The MVP will demonstrate an end-to-end AI-assisted meal analysis pipeline, from a smartphone meal image to personalised nutritional insights. The objective is to validate technical feasibility using a scaled-down dataset and controlled test scenarios, rather than deliver a production-ready medical application.

In Scope

Capability	***MVP Scope***
Meal Capture	*Capture/upload a meal image through a simple application interface*
Image Pre-processing	*Resize, normalize and prepare images for inference*
Food Instance Detection	*Detect and separate individual food/beverage items using* YOLO-Seg or Mask R-CNN
Food Classification	*Classify detected food instances using a* pre-trained CNN fine-tuned on FoodSG/Food-101
Portion Estimation	*Estimate approximate portion/volume of detected food instances*
Nutrition Mapping	*Map identified foods and estimated portions to a nutrition dataset for calories, carbohydrates, protein, fat, sugar, etc.*

Meal Aggregation	*Combine individual food instances into a complete meal-level nutritional profile*
User Context	*Maintain a simplified user profile: general/default, diabetes management, weight loss/gain, or fitness goals*
AI Contextualisation	*Combine meal nutrition and user context, with an* LLM used for explanation/recommendation generation
Recommendations	*Generate simple, actionable and explainable meal insights*
Application Output	*Present identified foods, estimated nutrition and recommendations back to the user*

MVP Validation

The MVP will evaluate the pipeline primarily on food detection/segmentation accuracy, food classification accuracy, nutrition-estimation accuracy, end-to-end response time, and quality/relevance of generated recommendations.

The primary use case will focus on Type 1 and Type 2 diabetes, particularly carbohydrate and portion awareness. Other health profiles will demonstrate the extensibility of the architecture.

Out of Scope

The MVP will not provide medical diagnosis, prescribe medication, recommend insulin dosage, replace professional dietary/medical advice, or claim clinical-grade nutritional accuracy. Continuous glucose monitoring integration, wearable integration, clinical validation, large-scale production deployment, regulatory certification and full ASEAN food coverage are reserved for future phases.

Data Collection and Preparation

For input and training data, use publicly available food-image datasets such as the NUS FoodLG and/or Kaggle's Food Nutrion datasets, supplemented by own meal photos to introduce real-world variation.

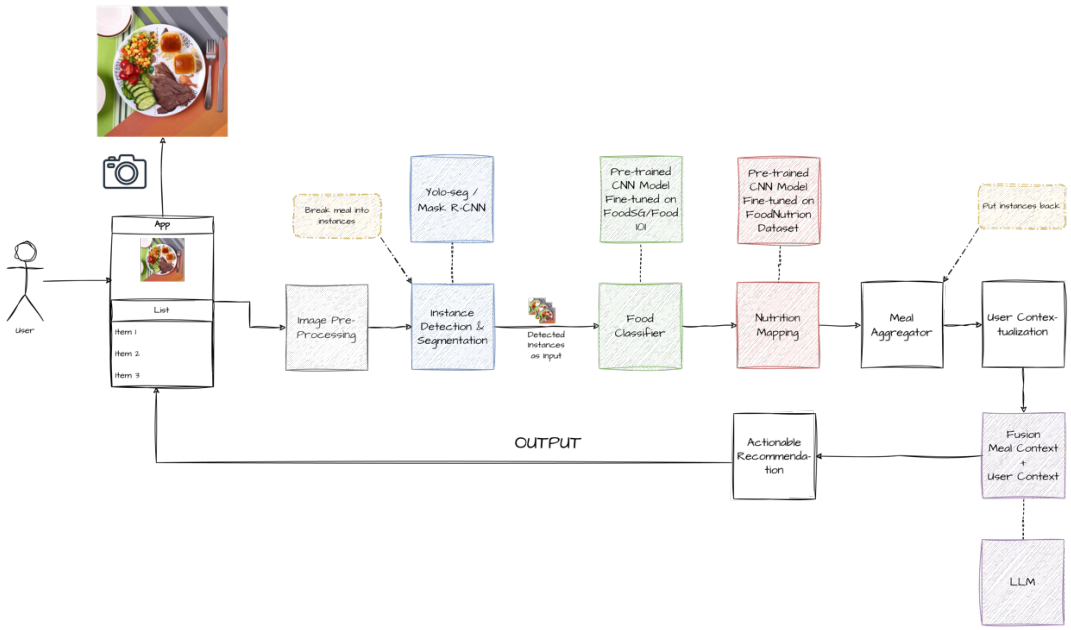
For output labels, I will map identified foods to nutrition information from the Kaggle Food Nutrition Dataset.

To improve robustness, I will include different lighting conditions, camera angles, portion sizes, backgrounds, cuisines, and visually similar dishes, then use data augmentation and separate training, validation, and test sets.

Training Datasets:
https://www.kaggle.com/datasets/utsavdey1410/food-nutrition-datasetLinks to an external site.

Localized version of Food 101
https://foodlg.comp.nus.edu.sg/download/zomvopbmiismeklwdhcb

Food dataset containing 101,000 images of food, organized by type of food. This is the basis of FoodSG dataset.
https://www.kaggle.com/datasets/dansbecker/food-101Links to an external site.

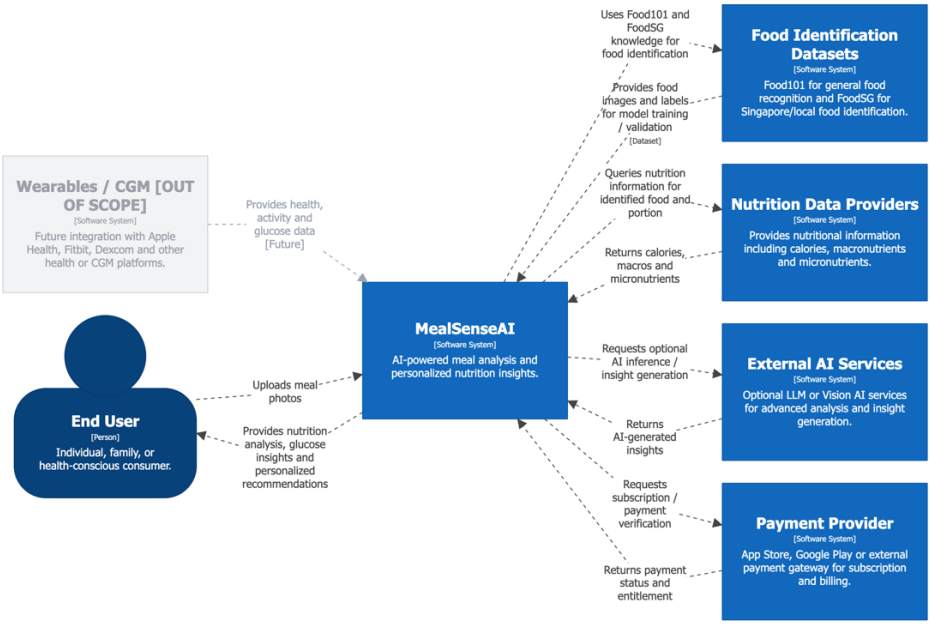


How much Data is needed:

For MVP project: ~5,000–10,000 labelled food images
For better MVP project: ~20,000–50,000 images
NUS' FoodSg-233 Dataset is consist of 209,861. This includes 233 Singapore foods from 13 food groups.

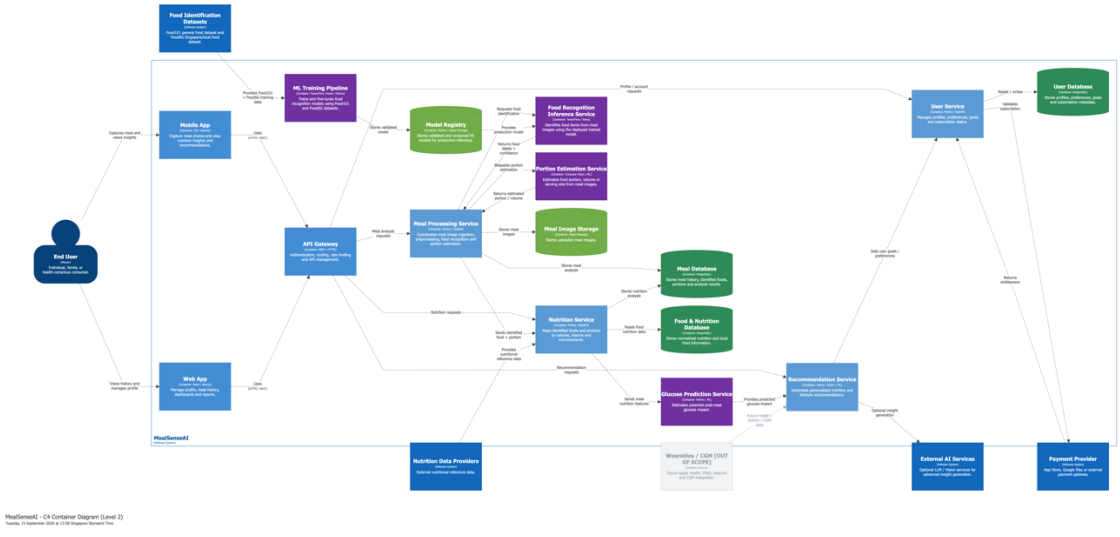
System Design

Context Diagram



MealSenseAI - C4 System Context (Level 1)
Tuesday, 15 September 2026 at 13:03 Singapore Standard Time

Container Diagram



MealSenseAI - C4 Container Diagram (Level 2)
Tuesday, 15 September 2026 at 13:08 Singapore Standard Time

Implementation (if available)

The project is currently under proposal review and pending approval.

Results and Progress (if available)

The project is currently under proposal review and pending approval.

Challenges and Roadblocks (if available)

The project is currently under proposal review and pending approval.

Future Work (if available)

The project is currently under proposal review and pending approval.

Conclusion

MealSenseAI addresses a growing need for accessible, AI-assisted nutrition and meal analysis, with an opportunity to combine food recognition, portion estimation, nutritional insights, and personalized recommendations with an ASEAN focus.

There is also a personal irony in developing this as someone living with Type 2 diabetes the tool intended to help manage health should not become another burden on the cost of living with the condition. This reinforces affordability as a core design principle as highlighted in the Market Analysis.

The market and technical assessment suggests that current AI/ML technologies are ready to validate the concept through an MVP. The next phase will focus on implementation, accuracy, usability, and proving that MealSenseAI can deliver useful health insights at an accessible price.

Supplementary Materials

AI Tool Declaration:
I used ChatGPT 5.6 Sol and Google's AI Mode for research, brainstorming, content refinement, technical exploration, and visual/content generation. I am responsible for project concept, objectives, market research, architecture, design decisions, validation, analysis, document review. I retain responsibility for the accuracy, decisions, and final work submitted.

Acknowledgement:
Thanks to Prof. Zhu Fangming for giving me the opportunity and flexibility to develop MealSenseAI as an individual project.

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